

**All design decisions made during our project, sequentially.****1. Microcontroller**

Our choice of micro controller was one of the most important decisions to the way the rest of the components were chosen. There were many factors to consider, mostly under the umbrella of software and hardware. Ultimately, our choice came down to two options, the Arduino Uno/Nano with the ATMEGA3P, and the Pi Zero W (for the form factor). Given considerations of difficulty of hardware configuration, as well as the fact that we had access to a Pi Zero W and GSM/GPRS hat already, we chose the Pi Zero W.

**2. GSM/GPRS Hat**

This was a choice that seemed easy, given the fact that we already had access to a SIMCOM R800C in a hat form factor for the Pi Zero, but ultimately configuration of the SIMCOM R800C, both in hat and USB form factor proved too difficult for reasons outlined in the Project Logbook. Given our failure to successfully configure it for the prototype, we switched gears and used the Sierra Wireless EM7455 WWAN interface connected to two adapter boards, one to adapt M.2 to miniPCIe and one to adapt miniPCIe to USB.

**3. GPS Module**

Our choice for GPS module was based on ease of use, as we wanted to implement the task of pulling coordinates in a Python script, so a GPS with a well documented library would be nice. Our choice came down to the SAM-M8Q which the professor graciously offered us, and the Adafruit Ultimate GPS, which was available at our local Micro Center. We decided to try and play both sides, in the hopes of coming out on top. We ultimately ended up using the Adafruit GPS for its vast Python library, allowing us to directly integrate core functions into an easy to use Python script.

**4. Software choices**

There aren't many software choices in our project, so I'm going to summarize them here. We chose Python as the language for our script because of familiarity, primarily, and pre-existing libraries. For a network management software for the WWAN interface, we chose NetworkManager, rather than ModemManager or configuring the device through AT commands because that is what worked. Every other method proved too convoluted or ineffective. Finally, the use of a remote software to log the location of the device devolved from using Nextcloud Phonetrack hosted remotely, to using an automated email as a proof of concept.

**5. Battery choices**

We wanted to integrate some type of battery in the device, just as a proof on concept, however we were unable to do so due to time constraints.